coursera | IBM

Advanced RAG with Vector Databases and Retrievers ✕

Today's goals

Complete any 3 learning items · 3/3
Watch 2 videos · 2/2
Complete a graded assessment

Today's Skill Points 120 XP
See skill progress

Module 1
Advanced Retrievers for RAG

Welcome to the Course

- ✓ Course Introduction
  Video • 4 min
- ○ **Course Overview**
  Reading • 5 min
- ✓ RAG and Agentic AI Professional Certificate Overview
  Video • 5 min
- ✓ Reading: Helpful Tips for Course Completion
  Ungraded Plugin • 5 min

Work with Advanced Retrievers in LangChain

- ○ Explore Advanced Retrievers in Langchain: Part 1
  Video • 2 min
- ○ Explore Advanced Retrievers in Langchain - Part 2
  Video • 5 min
- ○ Lab: Build a Smarter Search with LangChain Context Retrieval
  Ungraded App Item • 1h
- ○ [Optional] Interactive LangChain Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ○ Practice Quiz: Work with Advanced Retrievers in LangChain
  Practice Assignment • 14 min

Work with Advanced Retrievers in LlamaIndex

- ○ Advanced Retrievers in LlamaIndex
  Video • 8 min
- ○ Lab: Explore Advanced Retrievers in LlamaIndex
  Ungraded App Item • 1h
- ○ [Optional] Interactive LlamaIndex Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ○ Practice Quiz: Work with Advanced Retrievers in LlamaIndex
  Practice Assignment • 14 min

Module Summary and Evaluation

- ○ Summary and Highlights: Advanced Retrievers for RAG
  Reading • 3 min
- ○ Reading: Cheat Sheet: Advanced Retrievers for RAG
  Ungraded Plugin • 15 min
- ○ Graded Quiz: Advanced Retrievers for RAG
  Graded Assignment • 21 min
- ○ [Optional] Meet and Greet
  Discussion Prompt • 2 min

Module 2
Build a Comprehensive RAG Application

Course Overview

8 XP • Advanced RAG System Implementation

0:03 / 3:27 1x

Hello, and welcome to the **Advanced RAG with Vector Databases and Retrievers** course.

This course empowers learners to unlock the full potential of Retrieval-Augmented Generation (RAG) by mastering advanced retrieval techniques and vector database integration. Focused on practical, real-world application, the course delves into the powerful retrievers in LlamaIndex and LangChain, and also introduces Facebook AI Similarity Search (FAISS) as a core component of scalable vector search.

Through hands-on projects, learners will explore advanced retrieval patterns, integrate FAISS into RAG pipelines, and develop interactive AI applications using Gradio. Emphasis is placed on enhancing response accuracy in large language models (LLMs) through sophisticated retrieval strategies.

By the end of the course, learners will possess the skills to design, build, and deploy intelligent, production-ready RAG systems—preparing them to deliver high-performing AI-driven solutions across diverse domains.

This course is part of the IBM RAG and Agentic AI Professional Certificate.

Prerequisites

This course is accessible to a wide range of professionals but having some knowledge and experience of the Python programming language is required. Additional working knowledge of vector databases, RAG, and Similarity Search is advantageous.

It is recommended that you complete the following prior courses in the program before starting this one:

- Develop Generative AI Applications: Get Started
- Build RAG Applications: Get Started
- Vector Databases for RAG: An Introduction

Objectives

After completing this course, you will be able to:

- Build RAG applications using vector databases and advanced retrieval patterns
- Employ the core mechanics of vector databases such as FAISS and Chroma DB and implement indexing algorithms like HNSW
- Implement advanced retrievers using LlamaIndex and LangChain to improve the quality LLM responses
- Develop comprehensive RAG applications by integrating LangChain, FAISS, and front-end user interfaces built using Gradio

Course Outline

This course consists of two modules.

Module1: Advanced Retrievers for RAG

- Lesson 0: Welcome
- Lesson 1: Work with Advanced Retrievers in LangChain
- Lesson 2: Work with Advanced Retrievers in LlamaIndex
- Lesson 3: Module Summary & Assessment

Key Topics:

- Concept of retrievers in LangChain
- Concept of the vector store-based retriever
- How the vector store-based retriever works
- Comparison of the different types of LangChain retrievers
- Use cases for different LangChain retriever types
- Comparison of the different index types in LlamaIndex
- Use cases for index types
- TD-IDF
- Comparison of core and advanced retrievers
- Fusion strategies supported by the Query Fusion Retriever
- Use cases for different LlamaIndex retriever types

Module 2: Build a Comprehensive RAG Application

- Lesson 1: Introduction to FAISS for RAG
- Lesson 2: Module Summary & Assessment

Key Topics:

- Concepts of FAISS and Chroma DB
- Key differences between FAISS and Chroma DB
- Index types used by FAISS
- Concepts of locality-sensitive hashing (LSH) and hierarchical navigable small world (HNSW)
- Extending FAISS with Milvus
- Use cases for FAISS, Chroma DB, and Milvus

Congratulations on taking this step to boost your career in AI with next-level retrieval techniques for intelligent search and summarization. We hope you enjoy your learning journey.

Mark as completed

Like Dislike Report an issue

Go to next item →